Rocktim Jyoti Das

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Education

University of Maryland, College Park *2025 - present*
PhD in Computer Science
Advisor: Prof. Dinesh Manocha

Indian Institute of Technology Delhi *2018 - 2022*
B.Tech in Electrical Engineering with Minor in Computer Science and Engineering

Work Experience

AMD Research and Advanced Development, Austin *June, 2026 - present*
Research Associate - PhD
Mentor: Dr. Madhu Srinivasan
Physically Grounded World Model.

Mohamed bin Zayed University of Artificial Intelligence, Abu Dhabi *June, 2023 - Aug, 2025*
Research Associate II
Advisors: Prof. Ivan Laptev & Prof. Preslav Nakov
Robotic Manipulation and Reasoning.

Data Analytics and Intelligence Research Group, IIT Delhi & IBM Research, India *June, 2022 - June, 2023*
Research Assistant
Advisor: Prof. Mausam
Goal-oriented Dialog with Large Language Models

Publications

Conference Publications

1. **SLAT-Phys: Fast Material Property Field Prediction from Structured 3D Latents**
   Rocktim Jyoti Das, Dinesh Manocha.
   under review.
2. **BLAZER: Bootstrapping LLM-based Manipulation Agents with Zero-Shot Data Generation**
   Rocktim Jyoti Das*, Harsh Singh*, Diana Turmakhan, Abdullah Sohail, Mingfei Han, Preslav Nakov, Fabio Pizzati, Ivan Laptev.
   under review.
3. **MALMM: Multi-Agent Large Language Models for Zero-Shot Robotics Manipulation**
   Harsh Singh*, **Rocktim Jyoti Das***, Mingfei Han, Preslav Nakov, Ivan Laptev.
   accepted in IROS 2025.
4. **MALT: Improving Reasoning with Multi-Agent LLM Training**
   Sumeet Ramesh Motwani, Chandler Smith, **Rocktim Jyoti Das**, Markian Rybchuk, Ronald Clark, Philip Torr, Fabio Pizzati, Ivan Laptev, Christian Schroeder de Witt
   accepted in COLM 2025, ICLR SSI-FM 2025.
5. **EXAMS-V: A Multi-Discipline Multilingual Multimodal Exam Benchmark for Evaluating Vision Language Models**
   Rocktim Jyoti Das, Simeon Emilov Hristov, Haonan Li, Dimitar Iliyanov Dimitrov, Ivan Koychev, Preslav Nakov.
   Accepted in ACL Main, 2024.
6. **Beyond Size: How Gradients Shape Pruning Decisions in Large Language Models**
   Rocktim Jyoti Das, Mingjie Sun, Liqun Ma, Zhiqiang Shen.
   ArXiv 2024.
7. **Synergizing In-context Learning with Hints for End-to-end Task-oriented Dialog Systems**
   Vishal Saley, **Rocktim Jyoti Das**, Dinesh Raghu, Mausam.
   Accepted in EMNLP Main, 2024.
8. **DKAF: KB Arbitration for Learning Task-Oriented Dialog Systems with Dialog-KB Inconsistencies**
   Vishal Saley, **Rocktim Jyoti Das**, Dinesh Raghu, Mausam.
   Accepted in ACL Findings, 2023.
9. **Exploring Distributional Shifts in Large Language Models for Code Analysis**
   Shushan Arakelyan, **Rocktim Jyoti Das**, Yi Mao, Xiang Ren.
   Accepted in EMNLP Main, 2023.

Theses

1. **Efficient Framework for Feature-Aware Graph Coarsening**.
   Rocktim Jyoti Das*, Aditi Khandelwal*, Sandeep Kumar, Jaydeva.
   Undergraduate Thesis, Electrical Engineering, IIT Delhi, 2021 - 22.

Awards and Honors

- **Dean's Fellowship** awarded to selected first year students at University of Maryland, College Park *2025*
- One of 12, of 20k applicants selected for **Google Predoctoral Research Program, 2023** *2023*
- **Rajiv Bambawale Award** for Best Undergrad Thesis Award in EE Dept., IIT Delhi. *2022*
- **Anundoram Borooah Award**, Government of Assam, India for academic excellence in high school. *2016*

Research Projects

Multi-sensory Object Physics Estimation *March 2026 – present*
Advisors: Prof. Dinesh Manocha and Prof. Ruohan Gao
Developing physically grounded method for object physics estimation using multisensory data such as vision, audio and tactile.

Fast Vision-Based Material Field Estimation *Nov 2025 – March, 2026*
Advisor: Prof. Dinesh Manocha
Designing feed-forward network to estimate material properties of 3D assets (e.g., density, Young's modulus, Poisson's ratio) for physics-based simulation.

Learning Dynamics Models for Deformable Objects (Course Project) *Oct 2025 – Dec, 2025*
Advisor: Prof. Ming C. Lin
Developed physical-ground digital twins by inferring material properties from real-world observation videos to enable accurate simulation and robotic manipulation of deformable objects.

Service

- *Teaching Assistantship:*
  - Introduction to Computer Systems (UMD) - *Instr. Larry Herman* *Spring 2026*
  - Introduction to Data Science (UMD) - *Instr. Maksym Morawski* *Fall 2025*
  - Natural Language Processing (IITD) - *Prof. Mausam* *Fall 2022*
  - Advanced Machine Learning (IITD) - *Prof. Sandeep Kumar & Prof. Jaydeva* *Spring 2022*
  - Physical Electronics (IITD) - *Prof. Debanjan Bhowmik* *Fall 2021*
- *Journal Reviewer:* IEEE Robotics and Automation Letters (RA-L, 2025)
- *Conference Reviewer:* ICLR 2025, ICLR 2026, ICRA 2026, Neurips 2026, CoRL 2026, AAAI 2027
- *Organizer:* *Multimodal Reasoning Task* for ImageCLEF 2025 - Multimedia Retrieval Track at CLEF.
- *Outreach:* Mathematics tutor for high-school students at National Association of Blind, RK Puram, New Delhi.

Skills

- **Programming:** C++, MATLAB, Python and LaTeX
- **Framework:** PyTorch, PyTorch Geometric
- **Software:** CoppeliaSim, PyRep, RLBench, Issac-Sim
- **Robot:** Franka Emika Research 3

References

- Dr. Ivan Laptev, Professor, Department of Computer Vision, MBZUAI, Email: ivan.laptev@mbzuai.ac.ae.
- Dr. Mausam, Professor, Computer Science and Engineering Department, IIT, Delhi, Email: mausam@cse.iitd.ac.in.
- Dr. Preslav Nakov, Professor, NLP Department, MBZUAI, Email: preslav.nakov@mbzuai.ac.ae.
- Dr. Fabio Pizzati, Postdoctoral Researcher, Department of Computer Vision, MBZUAI, Email: fabio.pizzati@inria.fr.
- Dr. Christian Schroeder de Witt, RAEng Research Fellow, University of Oxford, Email: cs@robots.ox.ac.uk.